

OpenStax Calculus Volume 3

Section 6.2 Exercises: Line Integrals

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1. True or False? Line integral $\int_C f(x, y) ds$ is equal to a definite integral if C is a smooth curve defined on $[a, b]$ and if function f is continuous on some region that contains curve C .
2. True or False? Vector functions $\mathbf{r}_1 = t\mathbf{i} + t^2\mathbf{j}$, $0 \leq t \leq 1$, and $\mathbf{r}_2 = (1 - t)\mathbf{i} + (1 - t)^2\mathbf{j}$, $0 \leq t \leq 1$, define the same oriented curve.
3. True or False? $\int_{-C} (P dx + Q dy) = \int_C (P dx - Q dy)$
4. True or False? A piecewise smooth curve C consists of a finite number of smooth curves that are joined together end to end.
5. True or False? If C is given by $x(t) = t$, $y(t) = t$, $0 \leq t \leq 1$, then $\int_C xy ds = \int_0^1 t^2 dt$.

For the following exercises, use a computer algebra system (CAS) to evaluate the line integrals over the indicated path.

6. [T] $\int_C (x + y) ds$ $C: x = t$, $y = (1 - t)$, $z = 0$ from $(0, 1, 0)$ to $(1, 0, 0)$
7. [T] $\int_C (x - y) ds$ $C: \mathbf{r}(t) = 4t\mathbf{i} + 3t\mathbf{j}$ when $0 \leq t \leq 2$
8. [T] $\int_C (x^2 + y^2 + z^2) ds$ $C: \mathbf{r}(t) = \sin t\mathbf{i} + \cos t\mathbf{j} + 8t\mathbf{k}$ when $0 \leq t \leq \frac{\pi}{2}$
9. [T] Evaluate $\int_C xy^4 ds$, where C is the right half of circle $x^2 + y^2 = 16$ and is traversed in the clockwise direction.
10. [T] Evaluate $\int_C 4x^3 ds$, where C is the line segment from $(-2, -1)$ to $(1, 2)$.

For the following exercises, find the work done.

11. Find the work done by vector field $\mathbf{F}(x, y, z) = x\mathbf{i} + 3xy\mathbf{j} - (x + z)\mathbf{k}$ on a particle moving along a line segment that goes from $(1, 4, 2)$ to $(0, 5, 1)$.
12. Find the work done by a person weighing 150 lb walking exactly one revolution up a circular, spiral staircase of radius 3 ft if the person rises 10 ft.
13. Find the work done by force field $\mathbf{F}(x, y, z) = -\frac{1}{2}x\mathbf{i} - \frac{1}{2}y\mathbf{j} + \frac{1}{4}\mathbf{k}$ on a particle as it moves along the helix $\mathbf{r}(t) = \cos t\mathbf{i} + \sin t\mathbf{j} + t\mathbf{k}$ from point $(1, 0, 0)$ to point $(-1, 0, 3\pi)$.
14. Find the work done by vector field $\mathbf{F}(x, y) = y\mathbf{i} + 2x\mathbf{j}$ in moving an object along path C , the straight line which joins points $(1, 0)$ and $(0, 1)$.

15. Find the work done by force $\mathbf{F}(x, y) = 2y\mathbf{i} + 3x\mathbf{j} + (x + y)\mathbf{k}$ in moving an object along curve $\mathbf{r}(t) = \cos(t)\mathbf{i} + \sin(t)\mathbf{j} + \frac{1}{6}\mathbf{k}$, where $0 \leq t \leq 2\pi$.
16. Find the mass of a wire in the shape of a circle of radius 2 centered at (3, 4) with linear mass density $\rho(x, y) = y^2$.

For the following exercises, evaluate the line integrals.

17. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$, where $\mathbf{F}(x, y) = -1\mathbf{j}$, and C is the part of the graph of $y = \frac{1}{2}x^3 - x$ from (2, 2) to (-2, -2).
18. Evaluate $\int_\gamma (x^2 + y^2 + z^2)^{-1} ds$, where γ is the helix $x = \cos t, y = \sin t, z = t, 0 \leq t \leq T$.
19. Evaluate $\int_C yz dx + xz dy + xy dz$ over the line segment from (1, 1, 1) to (3, 2, 0).
20. Let C be the line segment from point (0, 1, 1) to point (2, 2, 3). Evaluate line integral $\int_C y ds$.
21. [T] Use a computer algebra system to evaluate the line integral $\int_C y^2 dx + x dy$, where C is the arc of the parabola $x = 4 - y^2$ from (-5, -3) to (0, 2).
22. [T] Use a computer algebra system to evaluate the line integral $\int_C (x + 3y^2) dy$ over the path C given by $x = 2t, y = 10t$, where $0 \leq t \leq 1$.
23. [T] Use a CAS to evaluate line integral $\int_C xy dx + y dy$ over path C given by $x = 2t, y = 10t$, where $0 \leq t \leq 1$.
24. Evaluate line integral $\int_C (2x - y) dx + (x + 3y) dy$, where C lies along the x-axis from $x = 0$ to $x = 5$.
25. [T] Use a CAS to evaluate $\int_C \frac{y}{2x^2 - y^2} ds$, where C is $x = t, y = t, 1 \leq t \leq 5$.
26. [T] Use a CAS to evaluate $\int_C xy ds$, where C is $x = t^2, y = 4t, 0 \leq t \leq 1$.

In the following exercises, find the work done by force field F on an object moving along the indicated path.

27. $\mathbf{F}(x, y) = -x\mathbf{i} - 2y\mathbf{j}$ C: $y = x^3$ from (0, 0) to (2, 8)
28. $\mathbf{F}(x, y) = 2x\mathbf{i} + y\mathbf{j}$ C: counterclockwise around the triangle with vertices (0, 0), (1, 0), and (1, 1)
29. $\mathbf{F}(x, y, z) = x\mathbf{i} + y\mathbf{j} - 5z\mathbf{k}$ C: $\mathbf{r}(t) = 2\cos t\mathbf{i} + 2\sin t\mathbf{j} + t\mathbf{k}, 0 \leq t \leq 2\pi$
30. Let F be vector field $\mathbf{F}(x, y) = (y^2 + 2xe^y + 1)\mathbf{i} + (2xy + x^2e^y + 2y)\mathbf{j}$. Compute the work of integral $\int_C \mathbf{F} \cdot d\mathbf{r}$, where C is the path $\mathbf{r}(t) = \sin t\mathbf{i} + \cos t\mathbf{j}, 0 \leq t \leq \frac{\pi}{2}$.
31. Compute the work done by force $\mathbf{F}(x, y, z) = 2x\mathbf{i} + 3y\mathbf{j} - z\mathbf{k}$ along path $\mathbf{r}(t) = t\mathbf{i} + t^2\mathbf{j} + t^3\mathbf{k}$, where $0 \leq t \leq 1$.
32. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$, where $\mathbf{F}(x, y) = \frac{1}{x+y}\mathbf{i} + \frac{1}{x+y}\mathbf{j}$ and C is the segment of the unit circle going counterclockwise from (1, 0) to (0, 1).
33. Force $\mathbf{F}(x, y, z) = zy\mathbf{i} + x\mathbf{j} + z^2x\mathbf{k}$ acts on a particle that travels from the origin to point (1, 2, 3). Calculate the work done if the particle travels:
 - (a) along the path $(0, 0, 0) \rightarrow (1, 0, 0) \rightarrow (1, 2, 0) \rightarrow (1, 2, 3)$ along straight-line segments joining each pair of endpoints;

(b) along the straight line joining the initial and final points.

(c) Is the work the same along the two paths?

[Figure: A curve and vector field in three dimensions. The curve segments go from $(1,2,3)$ to $(0,0,0)$ to $(1,0,0)$ to $(1,2,0)$, and the arrowheads point to $(0,0,0)$, $(1,0,0)$, and $(1,2,0)$. The surrounding vectors are larger the more the z component increases.]

34. Find the work done by vector field $\mathbf{F}(x, y, z) = x\mathbf{i} + 3xy\mathbf{j} - (x + z)\mathbf{k}$ on a particle moving along a line segment that goes from $(1, 4, 2)$ to $(0, 5, 1)$.
35. How much work is required to move an object in vector field $\mathbf{F}(x, y) = y\mathbf{i} + 3x\mathbf{j}$ along the upper part of ellipse $\frac{x^2}{4} + y^2 = 1$ from $(2, 0)$ to $(-2, 0)$?
36. A vector field is given by $\mathbf{F}(x, y) = (2x + 3y)\mathbf{i} + (3x + 2y)\mathbf{j}$. Evaluate the line integral of the field around a circle of unit radius traversed in a clockwise fashion.
37. Evaluate the line integral of scalar function xy along parabolic path $y = x^2$ connecting the origin to point $(1, 1)$.
38. Find $\int_C y^2 dx + (xy - x^2)dy$ along C: $y = 3x$ from $(0, 0)$ to $(1, 3)$.
39. Find $\int_C y^2 dx + (xy - x^2)dy$ along C: $y^2 = 9x$ from $(0, 0)$ to $(1, 3)$.

For the following exercises, use a CAS to evaluate the given line integrals.

40. [T] Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$, where $\mathbf{F}(x, y, z) = x^2z\mathbf{i} + 6y\mathbf{j} + yz^2\mathbf{k}$ and C is represented by $\mathbf{r}(t) = t\mathbf{i} + t^2\mathbf{j} + \ln t\mathbf{k}$, $1 \leq t \leq 3$.
41. [T] Evaluate line integral $\int_\gamma xe^y ds$ where, γ is the arc of curve $x = e^y$ from $(1, 0)$ to $(e, 1)$.
42. [T] Evaluate the integral $\int_\gamma xy^2 ds$, where γ is a triangle with vertices $(0, 1, 2)$, $(1, 0, 3)$, and $(0, -1, 0)$.
43. [T] Evaluate line integral $\int_\gamma (y^2 - xy) ds$, where γ is curve $y = \ln x$ from $(1, 0)$ toward $(e, 1)$.
44. [T] Evaluate line integral $\int_\gamma xy^4 ds$, where γ is the right half of circle $x^2 + y^2 = 16$.
45. [T] Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$, where $\mathbf{F}(x, y, z) = x^2y\mathbf{i} + (x - z)\mathbf{j} + xyz\mathbf{k}$ and C: $\mathbf{r}(t) = t\mathbf{i} + t^2\mathbf{j} + 2\mathbf{k}$, $0 \leq t \leq 1$.
46. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$, where $\mathbf{F}(x, y) = 2x \sin(y)\mathbf{i} + (x^2 \cos(y) - 3y^2)\mathbf{j}$ and C is any path from $(-1, 0)$ to $(5, 1)$.
47. Find the line integral of $\mathbf{F}(x, y, z) = 12x^2\mathbf{i} - 5xy\mathbf{j} + xz\mathbf{k}$ over path C defined by $y = x^2$, $z = x^3$ from point $(0, 0, 0)$ to point $(2, 4, 8)$.
48. Find the line integral of $\int_C (1 + x^2y) ds$, where C is ellipse $\mathbf{r}(t) = 2 \cos t\mathbf{i} + 3 \sin t\mathbf{j}$ from $0 \leq t \leq \pi$.

For the following exercises, find the flux.

49. Compute the flux of $\mathbf{F} = x^2\mathbf{i} + y\mathbf{j}$ across a line segment from $(0, 0)$ to $(1, 2)$.
50. Let $\mathbf{F} = 5\mathbf{i}$ and let C be curve $y = 0, 0 \leq x \leq 4$. Find the flux across C.
51. Let $\mathbf{F} = 5\mathbf{j}$ and let C be curve $y = 0, 0 \leq x \leq 4$. Find the flux across C.
52. Let $\mathbf{F} = -y\mathbf{i} + x\mathbf{j}$ and let C: $\mathbf{r}(t) = \cos t\mathbf{i} + \sin t\mathbf{j}$ ($0 \leq t \leq 2\pi$). Calculate the flux across C.

53. Let $\mathbf{F} = (x^2 + y^3)\mathbf{i} + (2xy)\mathbf{j}$. Calculate the flux of $\mathbf{F}$, oriented counterclockwise, across curve C : $x^2 + y^2 = 9$.
54. Find the line integral of $\int_C z^2 dx + ydy + 2ydz$, where C consists of two parts: C_1 and C_2 . C_1 is the intersection of cylinder $x^2 + y^2 = 16$ and plane $z = 3$ from $(0, 4, 3)$ to $(-4, 0, 3)$. C_2 is a line segment from $(-4, 0, 3)$ to $(0, 1, 5)$.
55. A spring is made of a thin wire twisted into the shape of a circular helix $x = 2 \cos t$, $y = 2 \sin t$, $z = t$. Find the mass of two turns of the spring if the wire has constant mass density.
56. A thin wire is bent into the shape of a semicircle of radius a . If the linear mass density at point P is directly proportional to its distance from the line through the endpoints, find the mass of the wire.
57. An object moves in force field $\mathbf{F}(x, y, z) = y^2\mathbf{i} + 2(x + 1)y\mathbf{j}$ counterclockwise from point $(2, 0)$ along elliptical path $x^2 + 4y^2 = 4$ to $(-2, 0)$, and back to point $(2, 0)$ along the x -axis. How much work is done by the force field on the object?
58. Find the work done when an object moves in force field $\mathbf{F}(x, y, z) = 2x\mathbf{i} - (x + z)\mathbf{j} + (y - x)\mathbf{k}$ along the path given by $\mathbf{r}(t) = t^2\mathbf{i} + (t^2 - t)\mathbf{j} + 3t\mathbf{k}$, $0 \leq t \leq 1$.
59. If an inverse force field $\mathbf{F}$ is given by $\mathbf{F}(x, y, z) = \frac{k}{\|\mathbf{r}\|^3}\mathbf{r}$, where k is a constant, find the work done by $\mathbf{F}$ as its point of application moves along the x -axis from $A(1, 0, 0)$ to $B(2, 0, 0)$.
60. David and Sandra plan to evaluate line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ along a path in the xy -plane from $(0, 0)$ to $(1, 1)$. The force field is $\mathbf{F}(x, y) = (x + 2y)\mathbf{i} + (-x + y^2)\mathbf{j}$. David chooses the path that runs along the x -axis from $(0, 0)$ to $(1, 0)$ and then runs along the vertical line $x = 1$ from $(1, 0)$ to the final point $(1, 1)$. Sandra chooses the direct path along the diagonal line $y = x$ from $(0, 0)$ to $(1, 1)$. Whose line integral is larger and by how much?